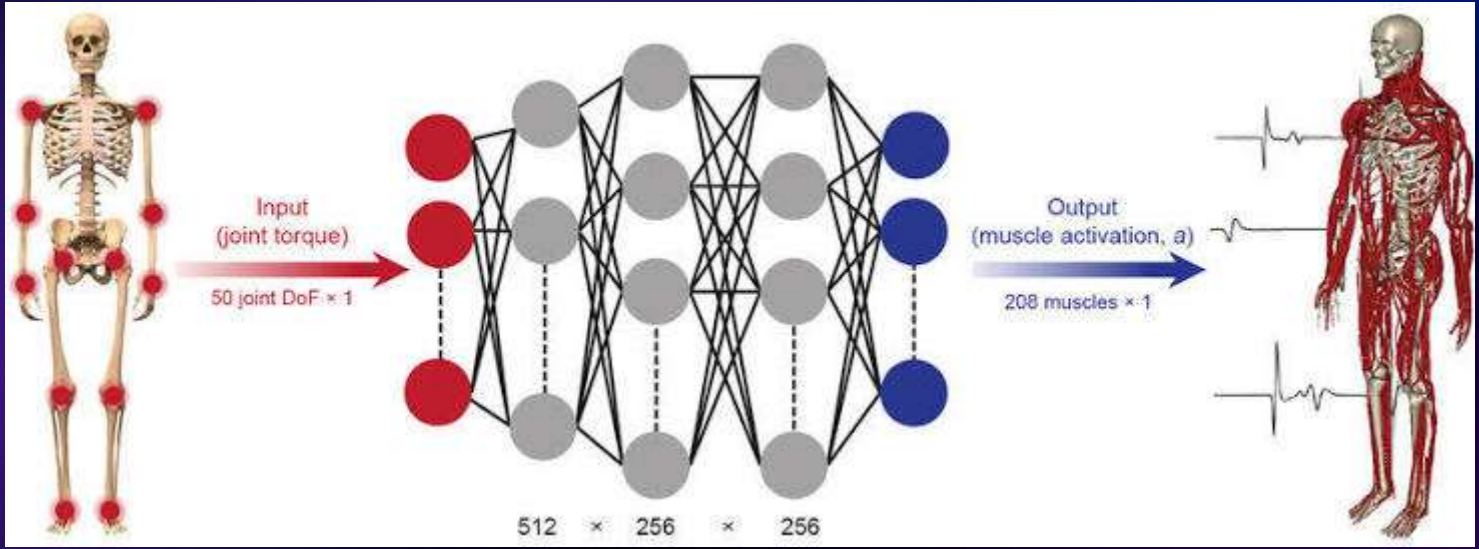
Presentation by Magdalena Dicheva, Mihaela Koseva and Nadya Radeva for the course 'Application of Mathematics In Modelling Real-life processes' At Faculty of Mathematics and Informatics, University 'St. Kliment Ohridski'

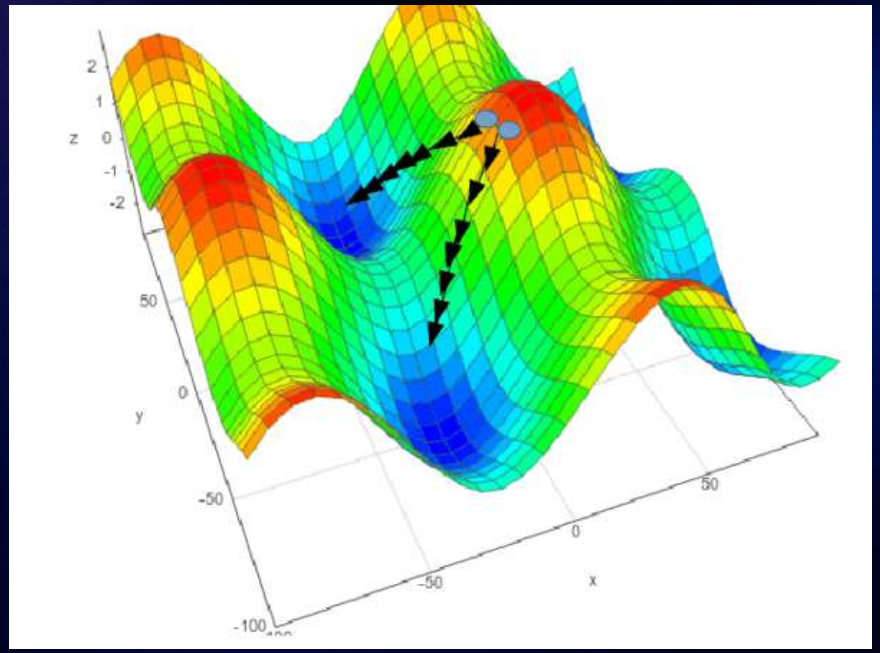
Neural Networks

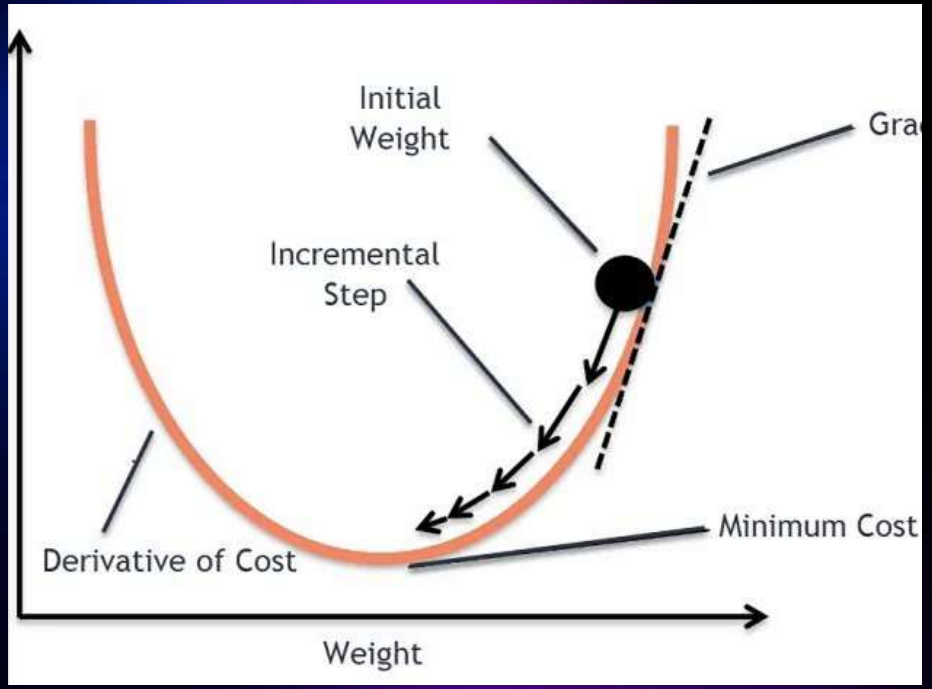
For modelling neural impulses

Neural Networks in the muscle activity



Back propagation and gradient descent





Backpropagation and Gradient Descent

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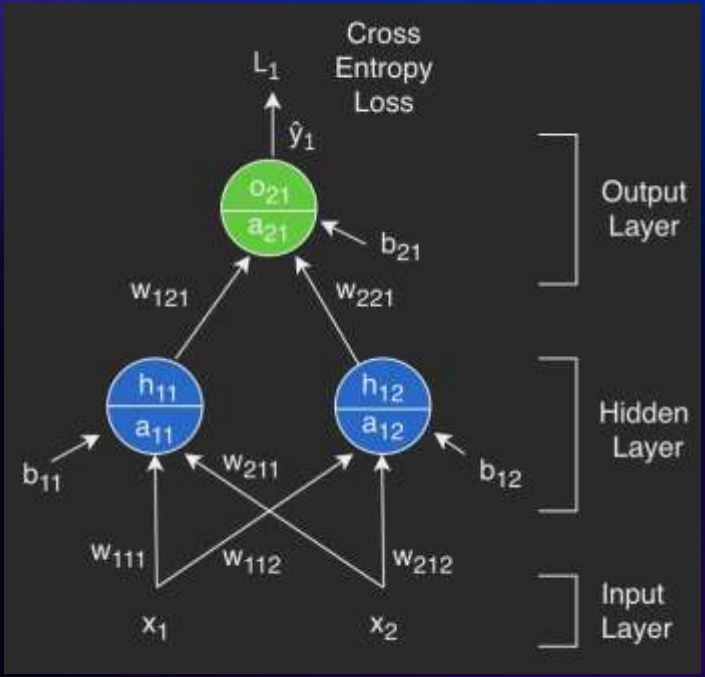
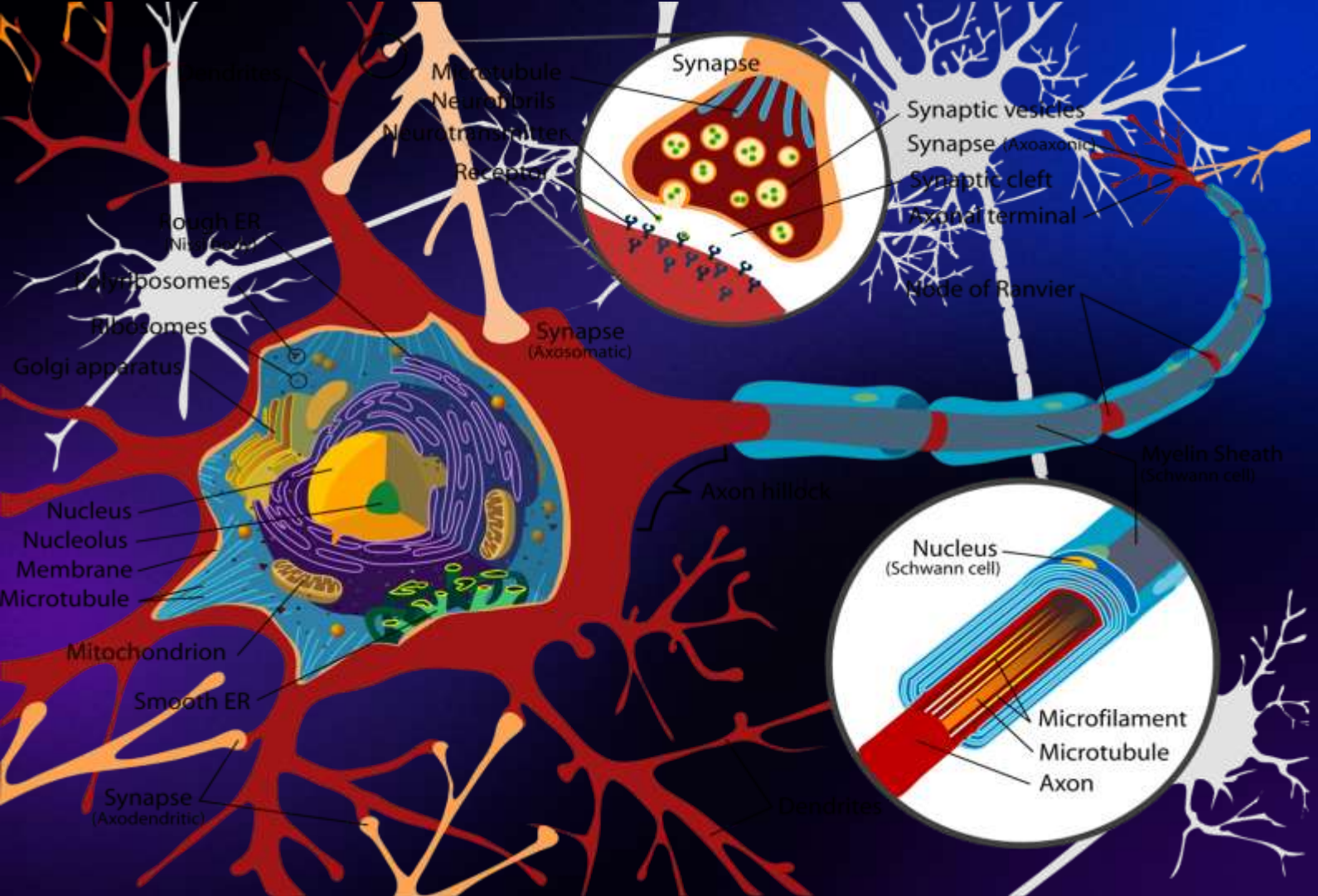
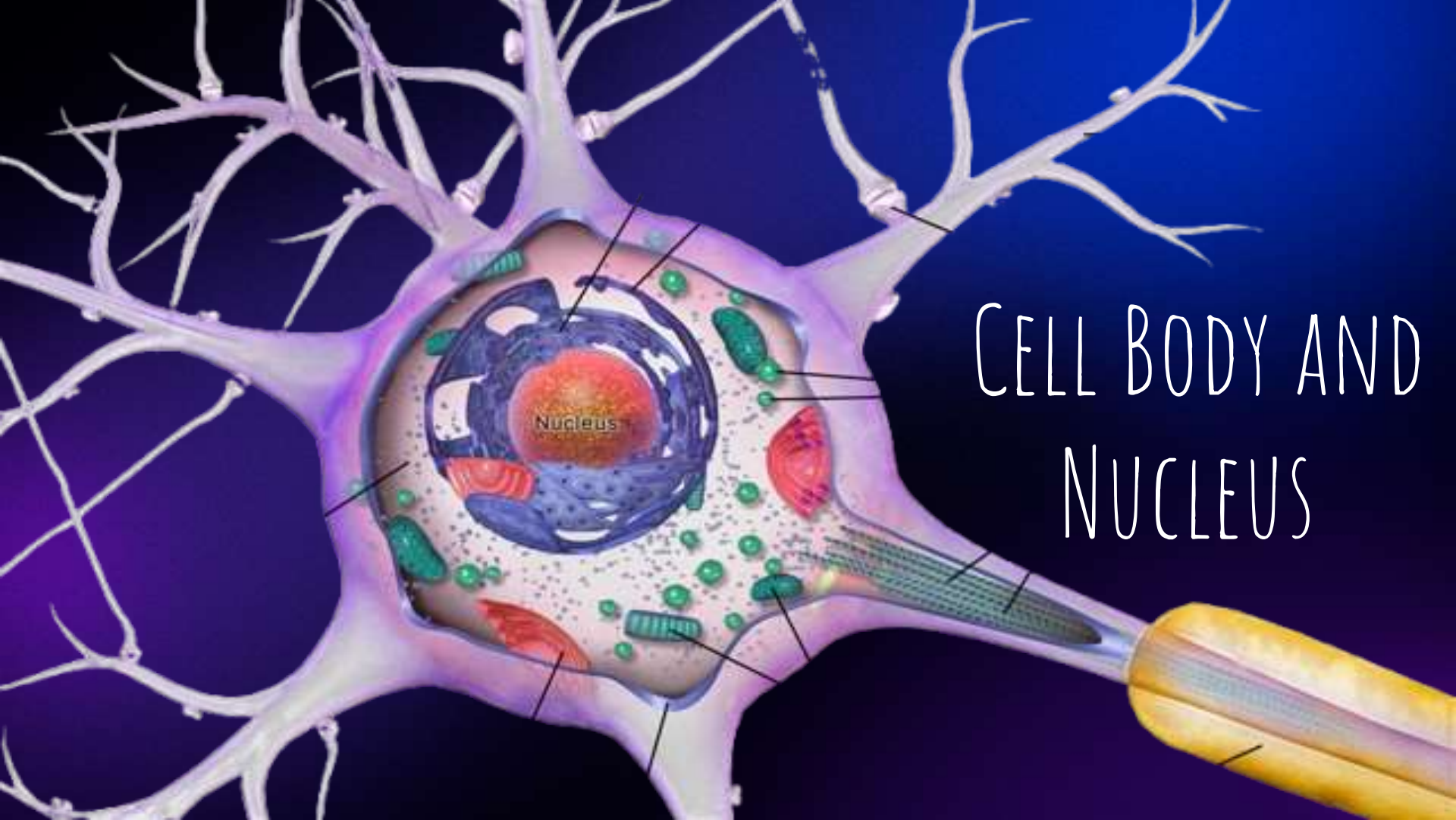
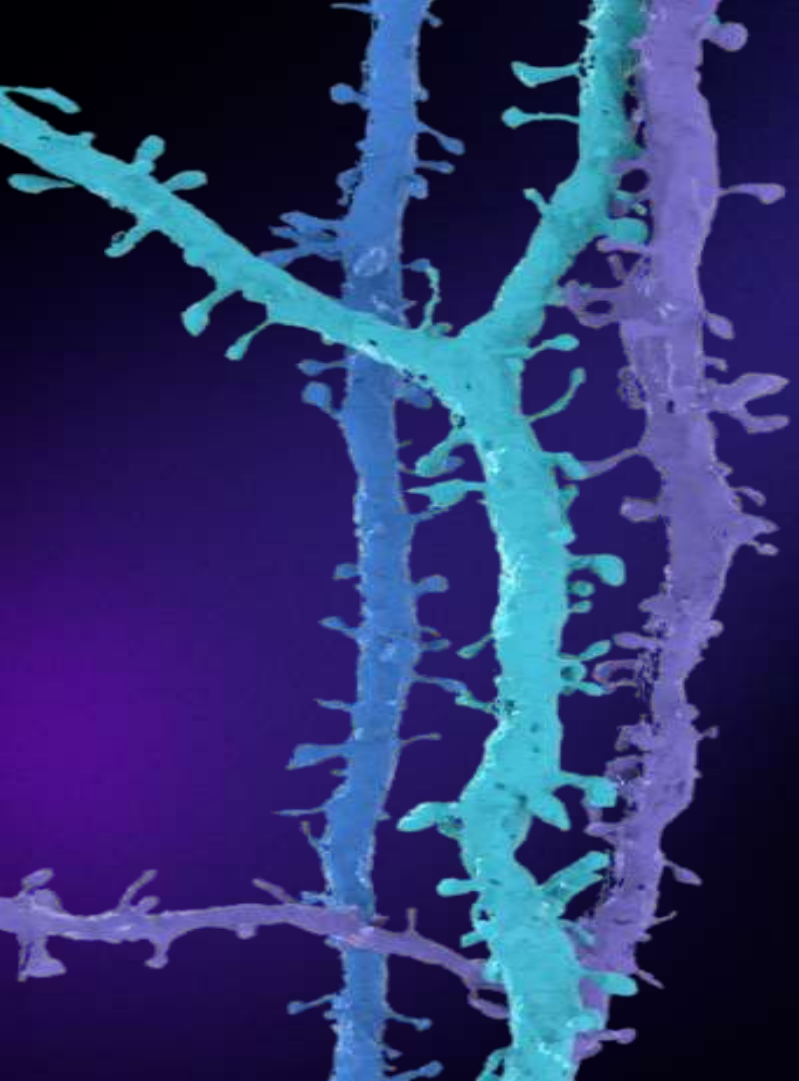


NEURONS


Dendrites
Microtubule
Neurofibrils
Neurotransmitter
Receptor
Synapse
Synaptic vesicles
Synapse (Axoaxonic)
Synaptic cleft
Axonal terminal
Rough ER
Nucleolus
Polyribosomes
Ribosomes
Golgi apparatus
Node of Ranvier
Synapse
(Axosomatic)
Myelin Sheath
(Schwann cell)
Axon hillock
Nucleus
Nucleolus
Membrane
Microtubule
Mitochondrion
Smooth ER
Synapse
(Axodendritic)
Dendrites
Nucleus
(Schwann cell)
Microfilament
Microtubule
Axon



Cell Body and Nucleus

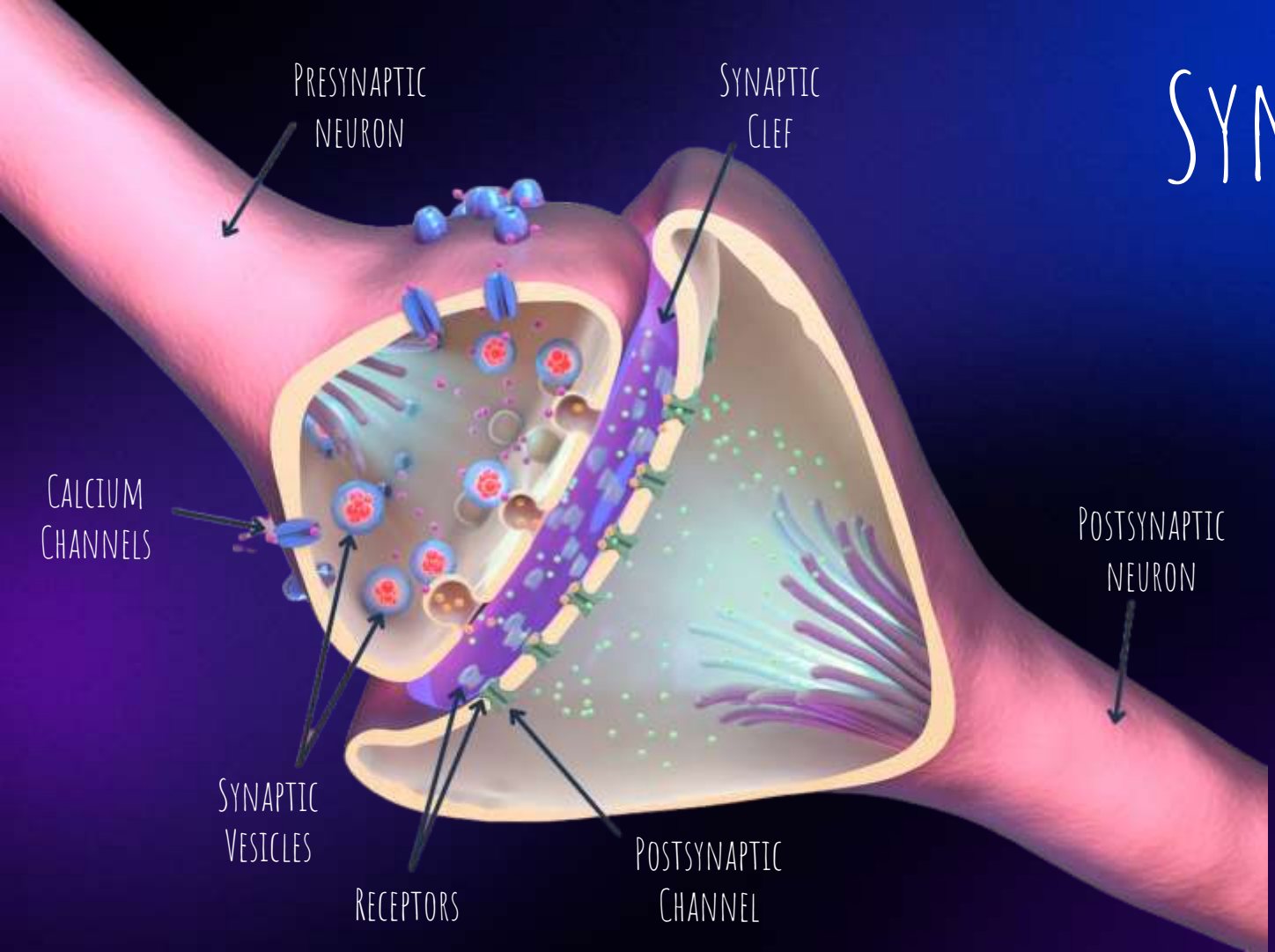


Dendrites

Dendrites are tree-like branches extending from the neuron's cell body that receive thousands of excitatory or inhibitory signals from other neurons, increasing the contact surface and determining—through signal summation—whether an action potential will be generated and sent along the axon.

Axon

The axon is the part of a neuron that carries electrical signals away from the cell body to other cells, allowing communication over long distances in the nervous system.

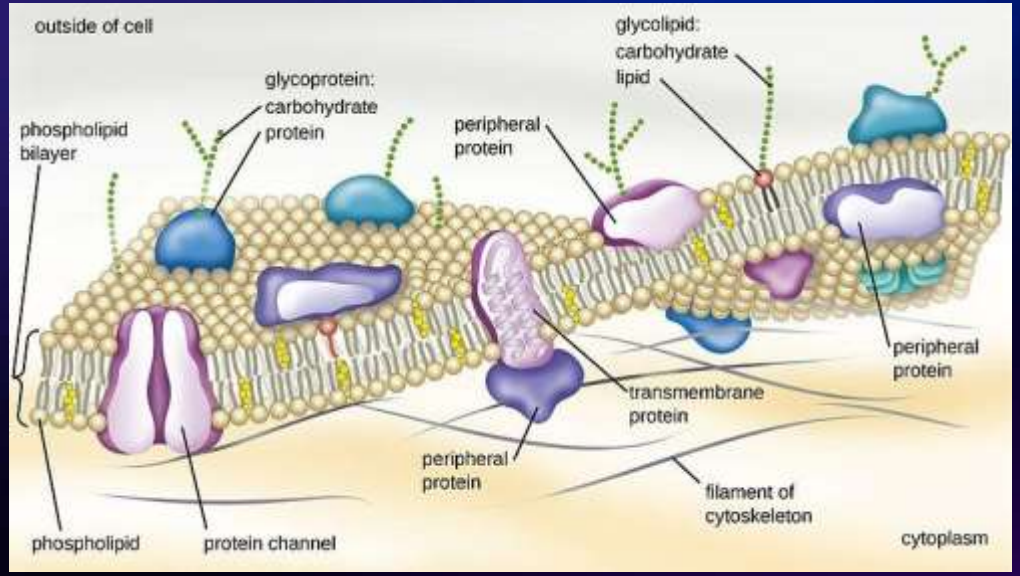


Synapse

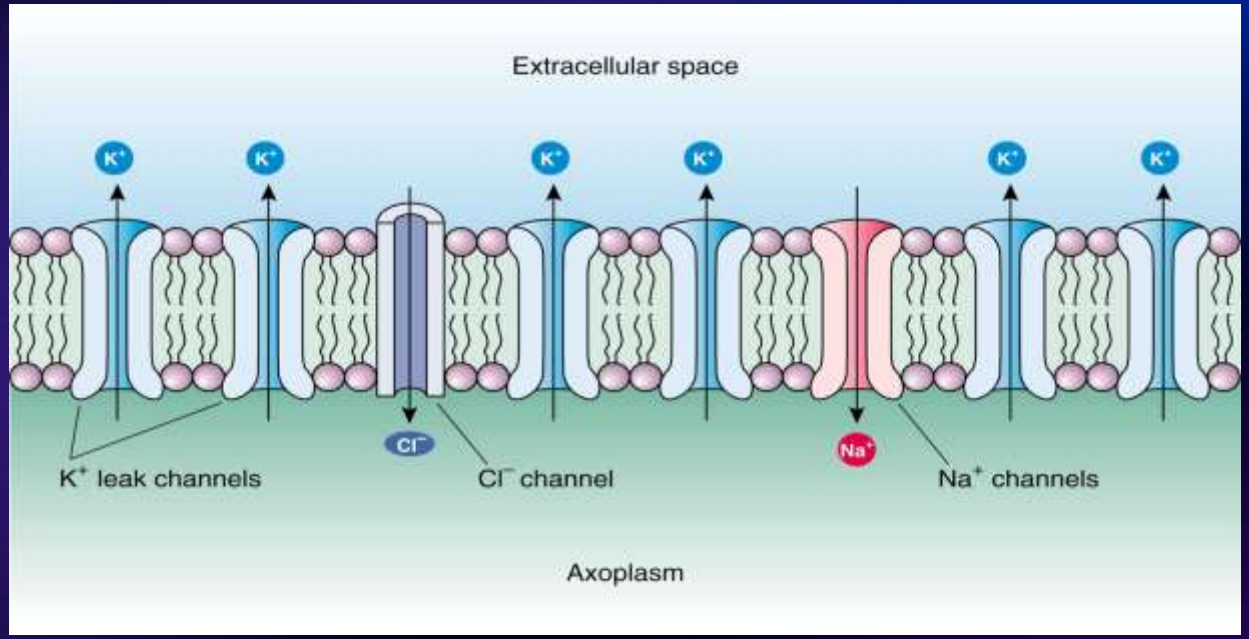
A synapse is a small junction between neurons where electrical signals are converted into chemical signals and transmitted to another cell, where they can either stimulate or inhibit its activity.

Neuronal membrane

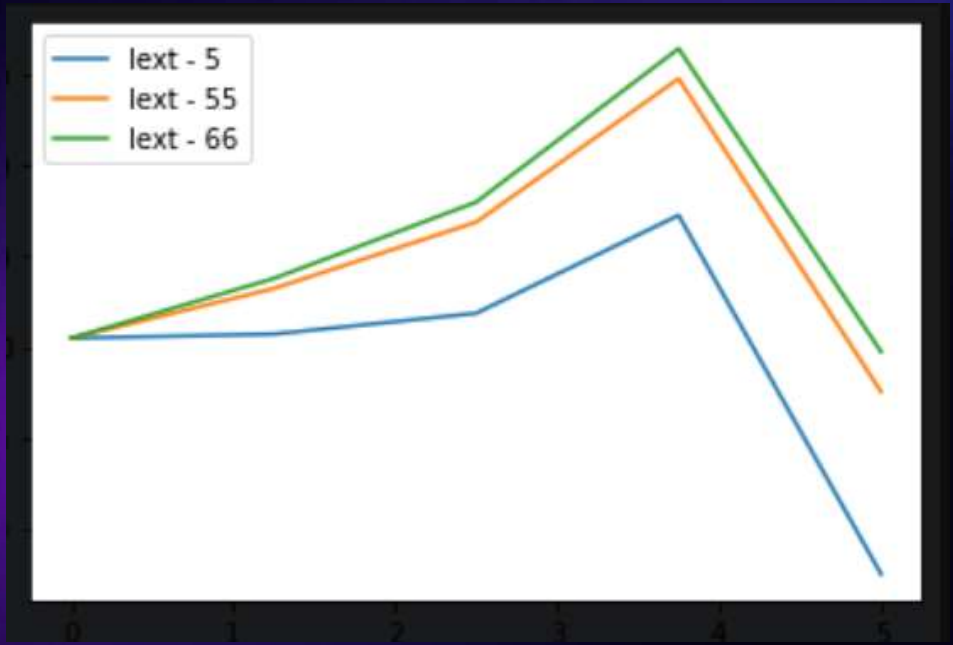
The neuronal membrane is a selectively permeable lipid-based barrier that regulates ion movement and maintains a resting electrical potential across the neuron, which is essential for generating and transmitting nerve impulses.

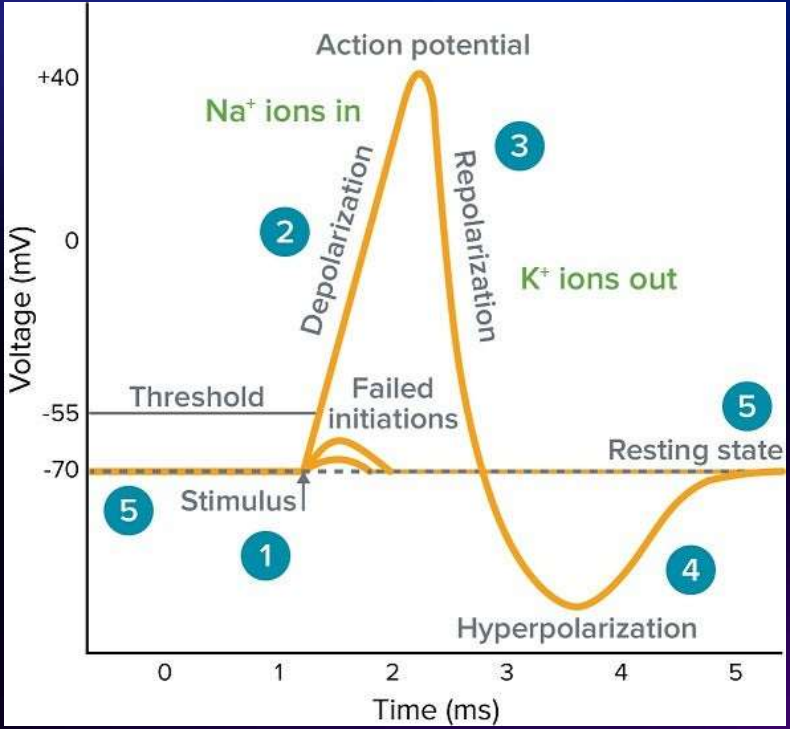


Activation potential



Neural Impulse





HH MODEL

$$I = C_m \frac{\mathrm{d}V_m}{\mathrm{d}t} + \bar{g}_{\mathrm{K}} n^4 (V_m - V_K) + \bar{g}_{\mathrm{Na}} m^3 h (V_m - V_{Na}) + \bar{g}_l (V_m - V_l),$$

$$\frac{dn}{dt} = \alpha_n(V_m)(1-n) - \beta_n(V_m)n$$

$$\frac{dm}{dt} = \alpha_m(V_m)(1-m) - \beta_m(V_m)m$$

$$\frac{dh}{dt} = \alpha_h(V_m)(1-h) - \beta_h(V_m)h$$

The End

We hope you enjoyed our presentation!